
PS Analysis in Data Science: Problem Set 3

Problem 1. Consider the curve $f : [0, 1] \rightarrow \mathbb{R}^2$ given by

$$f(t) = \begin{pmatrix} t \\ \frac{2}{3}t^{3/2} \end{pmatrix}$$

Determine the length of f .

Problem 2. Write Julia code to approximate the length of a curve $f : [0, 1] \rightarrow \mathbb{R}^2$ by implementing two approaches:

- (a) Given a partition $\mathcal{P}$ of $[0, 1]$, determine the polygonal length of f along $\mathcal{P}$.
- (b) Approximate the length integral by the mid-point rule.
- (c) Test and compare your code for several illustrative curves.

Problem 3. Consider the curve $f : [0, 1] \rightarrow \mathbb{R}^2$ given by

$$f(t) = \begin{pmatrix} t \cos(6\pi t) \\ t \sin(6\pi t) \end{pmatrix}$$

- (a) Determine the length of f by your best effort (how many digits are you able to determine accurately?).
- (b) Write a Julia function, whose input is a curve $\gamma : [0, 1] \rightarrow \mathbb{R}^2$ and a point $t_0 \in [0, 1]$ that returns an (affine) linear function that approximates γ in a neighborhood of t_0 .
- (c) Visualize the affine linear approximation for f and for several other different curves.

Problem 4. Consider the upper semicircle

$$f : [0, \pi] \rightarrow \mathbb{R}^2, \quad f(x) = \begin{pmatrix} \cos x \\ \sin x \end{pmatrix}$$

and $\varphi : [0, 1] \rightarrow [0, \pi]$ given by $\varphi(t) = \pi t^2$. Let

$$\gamma := f \circ \varphi.$$

- (a) Determine the metric derivative of γ .
- (b) Compute the length of γ .